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Triassic paleomagnetic data from south China and their bearing on the tectonic evolution of the western circum-Pacific region

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Abstract

We report Early and Middle Triassic paleomagnetic data from the south Chinese provinces of Fujian and Guangxi. The characteristic magnetization of the rocks in each case is concluded to be a primary remanence that passes the fold test. The Triassic pole from western Guangxi and four other Triassic and Late Permian poles from three provinces that lie on undisputed parts of the Yangtze craton are well clustered. This suggests that Guangxi (except the southeast part) was also a part of the Yangtze craton, at least since the Triassic and probably since the Late Permian. With respect to the Yangtze craton, between the Early Triassic and the Late Cretaceous, Fujian may have been rotated $121 \pm 9°$ counterclockwise and displaced $22 \pm 9°$ north, or rotated $59 \pm 9°$ clockwise and displaced $3 \pm 9°$ south, depending on the hemisphere in which the magnetization was acquired. Based on consistency with other paleomagnetic results, and in accordance with the geologic data from the area, the former interpretation is preferred. A remarkable coincidence of Mesozoic poles for south China and south Korea is observed which, if true, implies that the South China Block (except for the displaced terranes in the coastal provinces) and Korea may have been part of the same continental landmass from the Triassic onwards. This is consistent with some geologic observations suggesting affinities between the two places. It also implies that the major fault zones in north China (e.g., the Tan-Lu fault) are unrelated to major fault zones in southeast China (e.g., the Changle–Nanao fault), with the former probably associated with the suturing of north and south China and the latter influenced by proto-Pacific plate motion.

1. Introduction

Paleomagnetic research is beginning to elucidate the complex relationships between the tectonic blocks that comprise China (Fig. 1). Early investigations [1] found a discordance in the data for north and south China which suggested that the two were widely separated in the Late Permian and Early Triassic. Since then, more work has established a better understanding of the tectonic relationship between the two blocks. It is now believed that the landmasses collided by the

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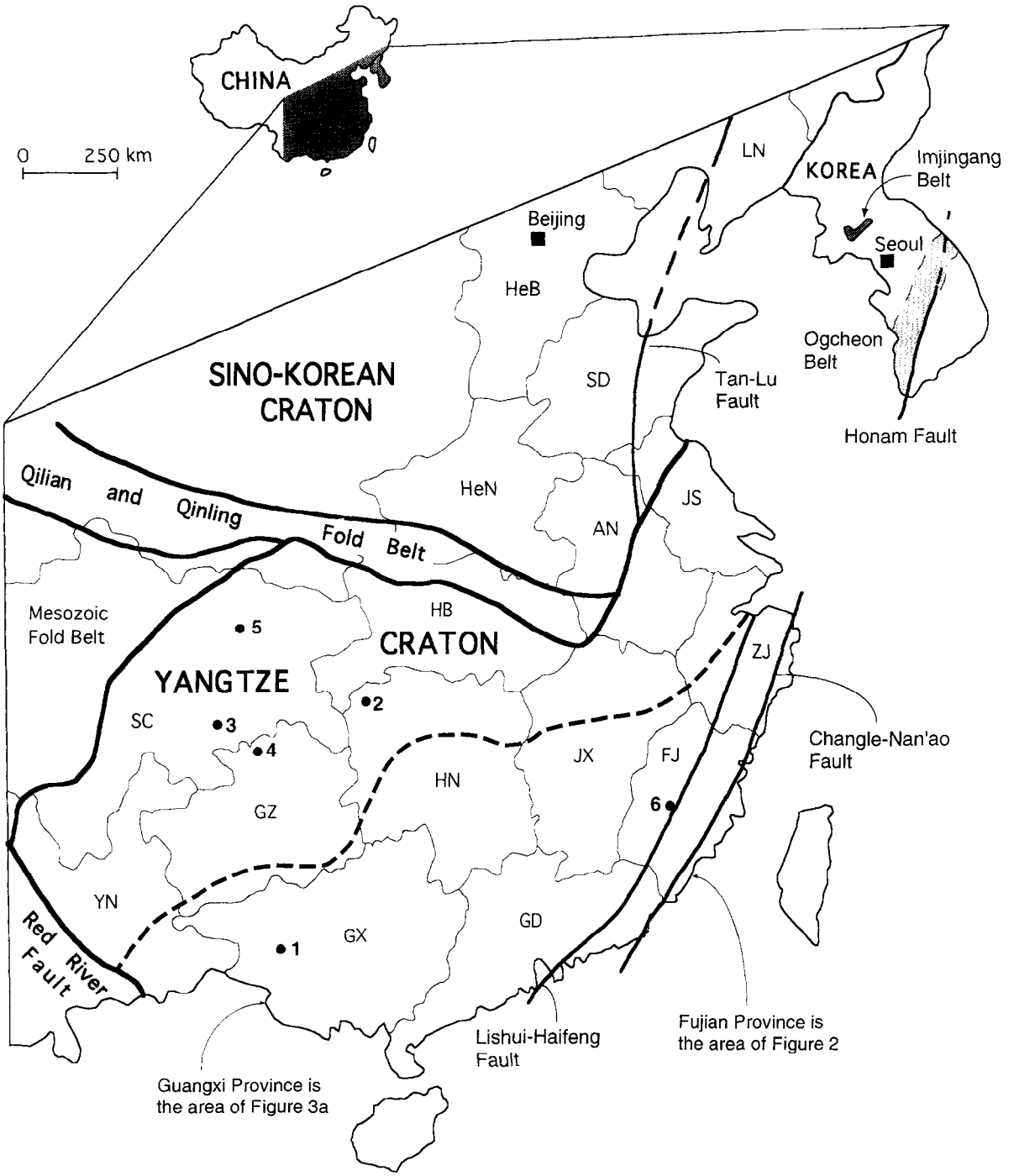


Fig. 1. Regional map of south and east China and Korea. Tectonic boundaries (heavy lines) are from Zhang et al. [6]. Dashed boundary delineates the Yangtze craton from the Huanan Block as proposed by Hsu et al. [4]. Numbers refer to the locations of the various studies discussed in the text (see Table 2). Locations 1 and 6 are from this study and the others are from the studies used to define the Yangtze craton reference pole. Provinces: *AN* = Anhui; *FJ* = Fujian; *GD* = Guangdong; *GZ* = Guizhou; *HeB* = Hebei; *HeN* = Henan; *HB* = Hubei; *HN* = Hunan; *JS* = Jiangsu; *JX* = Jiangxi; *LN* = Liaoning; *SC* = Sichuan; *SD* = Shandong; *YN* = Yunnan; *ZJ* = Zhejiang.

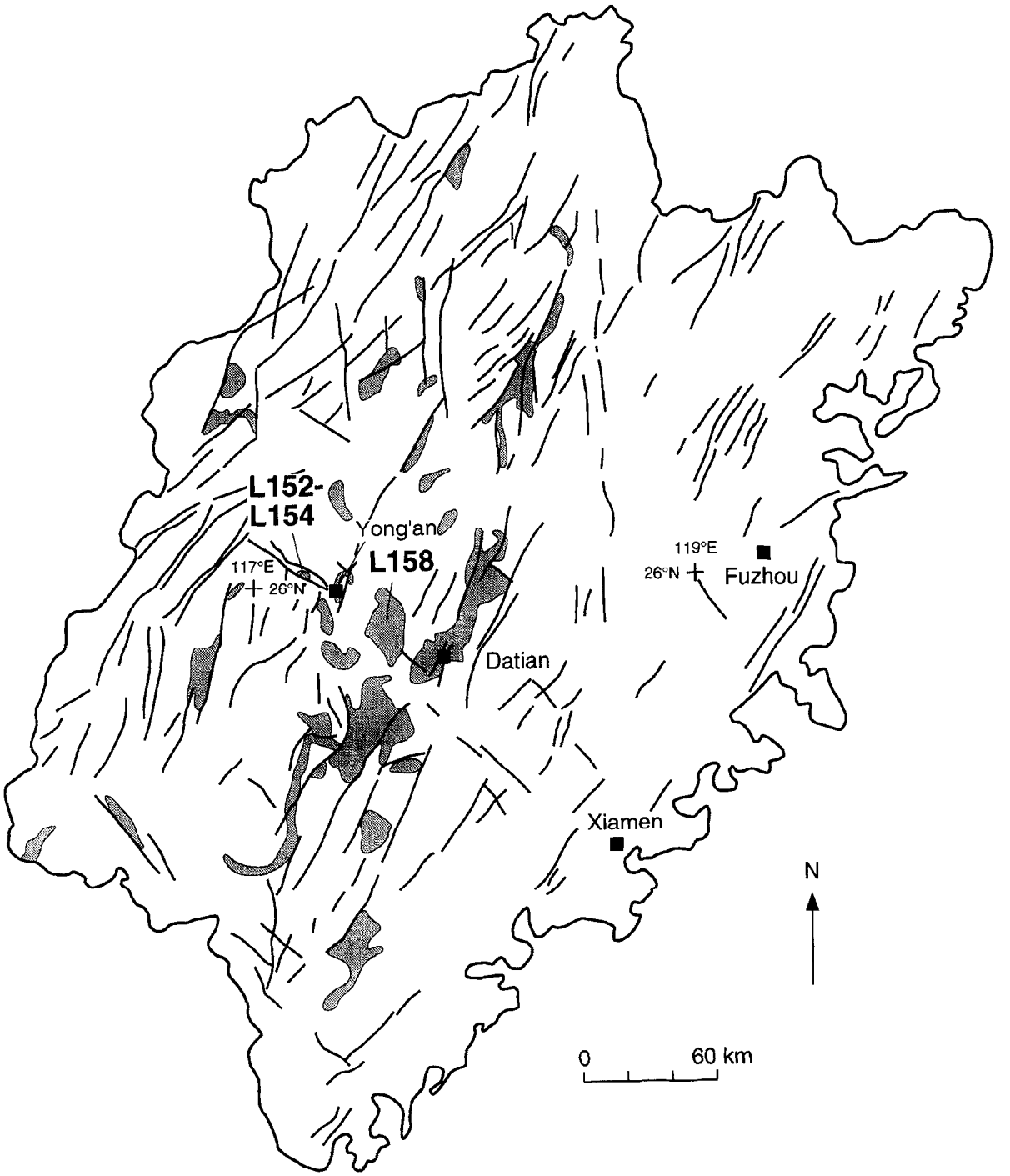
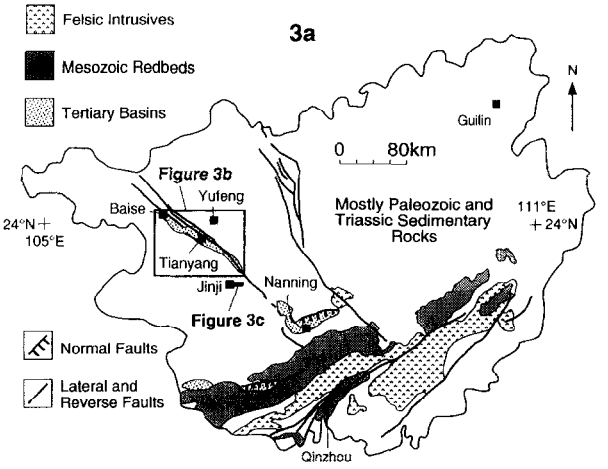
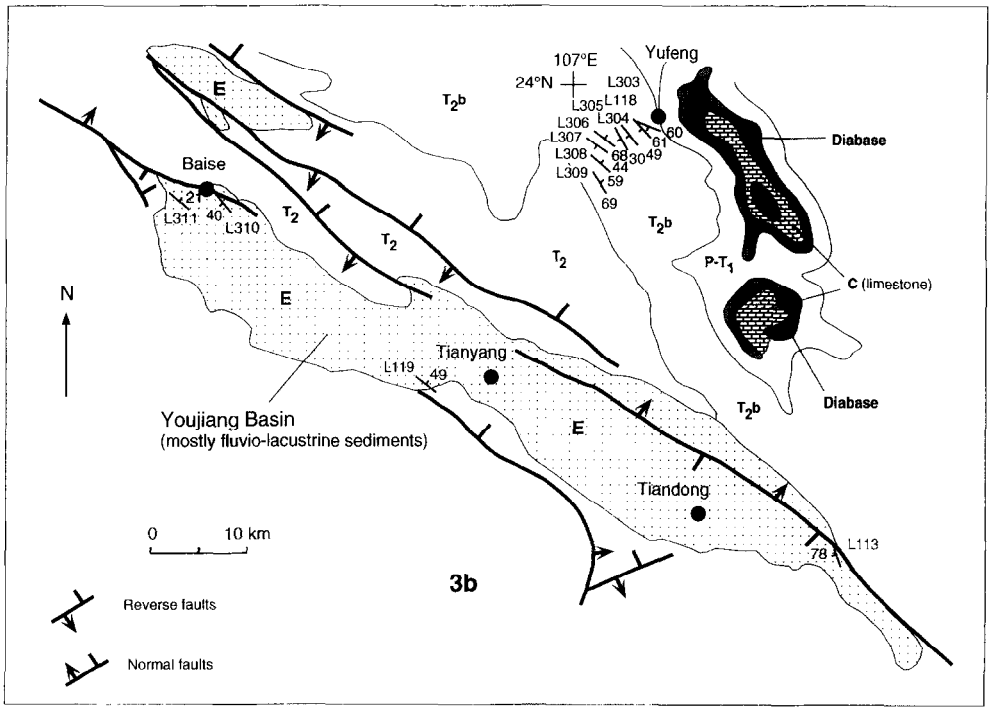
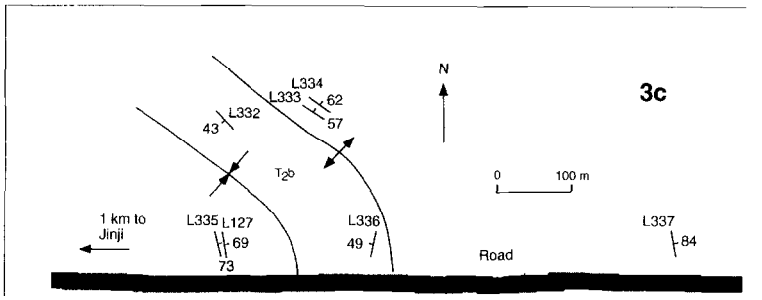


Fig. 2. Simplified geologic map of Fujian Province showing distribution of faults and Mesozoic sedimentary basins (from MGMR [9]). L152–L154 and L158 are the sample localities from the Early Triassic Xikou Formation (this study). No reinterpretation of the fault geometry depicted by MGMR [9] has been attempted.


Felsic Intrusives
Mesozoic Redbeds
Tertiary Basins
3a
N
Guilin
0 80km
Figure 3b
Yufeng
Baise
24°N+
105°E
Mostly Paleozoic and
Triassic Sedimentary
Rocks
111°E
+24°N
Tianyang
Jinji
Nanning
Figure 3c
Normal Faults
Lateral and
Reverse Faults
Qinzhou


107°E
24°N
Yufeng
L303
L305
L118
L306
L304
L307
L308
L309
60
61
68
30
49
44
59
69
Diabase
E
T2b
Baise
21
L311
40
L310
T2
T2
T2
T2b
P-T1
C (limestone)
N
E
Tianyang
L119
49
Youjiang Basin
(mostly fluvio-lacustrine sediments)
Diabase
T2b
E
Tiandong
0 10 km
78
L113
3b
Reverse faults
Normal faults


L334
L333
62
57
L332
43
N
3c
T2b
0 100 m
1 km to
Jinji
L335
L127
69
73
L336
49
Road
L337
84

Triassic and then scissored together [2]. The collision produced a suture represented today by the Qinling mountains.

The tectonic framework within south China, however, remains enigmatic because widespread overprinting often obscures paleomagnetic results (e.g., [3]). Tectonic interpretations based on geologic observations are also inconclusive. While most workers agree that the South China Block (SCB) is composed of tectonic terranes, the temporal and kinematic amalgamation history of these terranes is poorly understood. Large-scale overthrusts observed in Jiangxi and Anhui Province are interpreted by Hsu et al. [4] as evidence supporting the Mesozoic collision of landmasses, reminiscent of Alpine tectonics. Left-lateral offset of marker beds of the order of hundreds of kilometers and the identification of major shear zones have prompted Xu et al. [5] to propose that the entire east part of the Chinese landmass was dominated by a Mesozoic sinistral shear system.

In this paper we present Triassic paleomagnetic data that appear to have partially escaped the tenacious overprinting so prevalent in south China. The data are sufficient to begin addressing the ambiguities inherent in the tectonic evolution of the area. We then use these results, in concert with others, to explore other problems related to the development of the western circum-Pacific region.

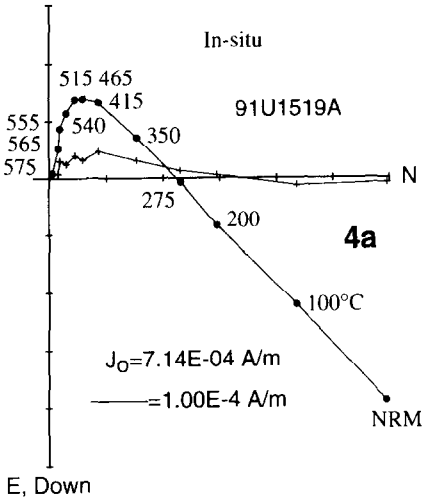
2. Geology and sampling

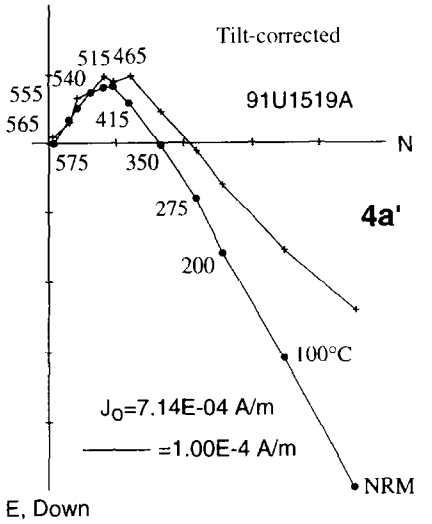
We sampled Early and Middle Triassic strata in southeast China at areas one (Guangxi Province) and six (Fujian Province) in Fig. 1. In Fujian Province, we drilled 32 cores at 4 localities in the Xikou Formation (Fig. 2). Two localities were sampled along a road that connects Yongan City with the Ansha reservoir, one locality was sampled at an outcrop along the reservoir, and the fourth locality was sampled along the road that connects Yongan and Datian. The formation is composed of gray to gray–green, medium- to thin-bedded calcareous and argillaceous siltstone and mudstone, and it attains 1500 m in thickness. Intercalation of thick-bedded or lens-like limestone occurs in the middle and lower part of the formation [7]. The identification of classical Bouma sequences and other sedimentary structures within the formation indicates turbidites deposited in a lower continental slope environment [8]. The pelycepods *Eumorphotis multiformis*, *E. inaequicostata*, *Entolium unionites* and *Claraia griesbachi* and the ammonite *Ophiceras* sp. found in the Xikou units indicate an Early Triassic age [8,10].

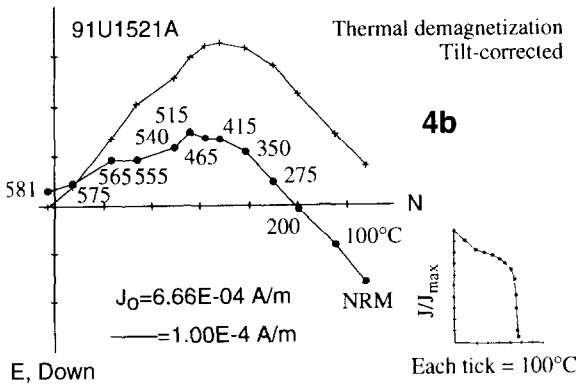
The Xikou strata are convoluted and in places are metamorphosed to hornfels-facies conditions. The faulting and deformation observed in Fujian (Fig. 2) is attributed to the northward translation of terranes during oblique subduction of oceanic crust under the Chinese continent [11]. Motion along some of the faults such as the left-lateral Changle–Nan'ao shear zone (Fig. 1) is Jurassic through Early Cretaceous in age [5,12].

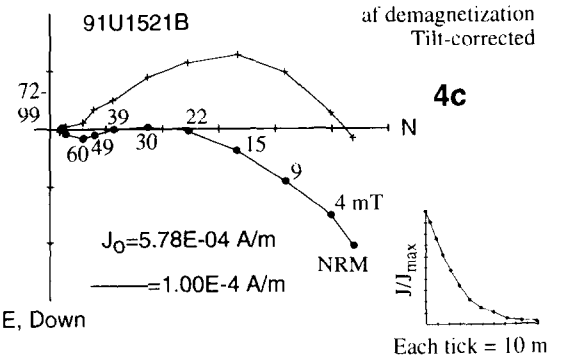
In western Guangxi Province (Fig. 3a), we drilled 123 cores at 15 localities in the Baifeng Formation. The samples were obtained from two transects, one along a 15 km stretch of road that connects Yufeng and Tianyang cities, and another a few kilometers west of Jinji. We refer to the transects as the Yufeng and Jinji sections respectively.

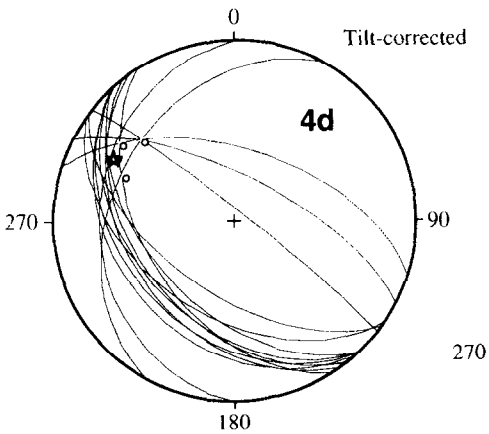
Fig. 3. Simplified geologic map of the area around Yufeng (b) and the Jinji section (c) in Guangxi Province (a) (from GXBGMR [13]). Because the trend of the Triassic folds in the Yufeng section (see (b)) parallels both the trend of the Youjiang fault system and the strike of the Early Tertiary sediments found in the fault zone, a similar age of deformation for all the rocks in the area is likely (e.g., Cenozoic). The folding in the Jinji area (c) is more complex. Attitudes in the north of the section parallel the Cenozoic trend while those in the south trend northerly. Heavy lines indicate faults, light lines indicate geologic contacts. C = Carboniferous; P = Permian; T_1 = Early Triassic; T_2 = Middle Triassic; T_2b = Middle Triassic Baifeng Formation; E = Cenozoic. Locality names next to strike/dip symbols begin with an L. Localities L113 and L119 are from Gilder et al. [14]; localities L310 and L311 are from unpublished data; the other localities are from this study.

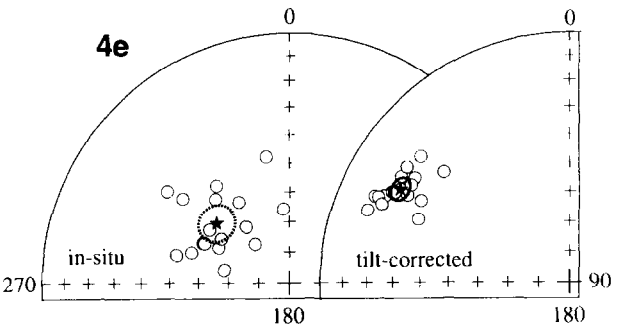

In-situ
91U1519A
515 465
415
555
565
575
540
350
N
275
200
4a
100°C
J_O=7.14E-04 A/m
=1.00E-4 A/m
NRM
E, Down


Tilt-corrected
515465
540
555
565
91U1519A
415
N
575
350
275
4a'
200
100°C
J_O=7.14E-04 A/m
=1.00E-4 A/m
NRM
E, Down


91U1521A
Thermal demagnetization
Tilt-corrected
4b
515
540
415
465
350
581
565
555
275
575
N
200
100°C
J_O=6.66E-04 A/m
NRM
=1.00E-4 A/m
J/J_{max}
E, Down
Each tick = 100°C


91U1521B
af demagnetization
Tilt-corrected
4c
72-
99
39
22
N
60
49
30
15
9
4 mT
J_O=5.78E-04 A/m
NRM
=1.00E-4 A/m
J/J_{max}
E, Down
Each tick = 10 m


0
Tilt-corrected
4d
270
90
180


4e
0
0
in-situ
tilt-corrected
270
90
180
180

The Baifeng Formation consists mostly of fine-grained, gray to gray–green sandstone, siltstone and shale; white to orange-red mudstones are also present. The ammonites *Anolcites* sp., *Balatonites* sp., *B. balatonicus*, *B. multisulcatus*, *Beyrichites* sp., *Bulogites* sp., *Cuccoceras* sp., *Danubites* sp., *D. kansa*, *Hollandites* sp., *Huishuites* sp., *Japonites* sp., *Judicarites* sp., *Leiophyllites* sp., *L.* cf. *confucii*, *Monophyllites* sp., *Paraceratites* sp., *Paracrochordiceras* sp. and *Sageceras* sp. and the pelycepods *Daonella producta*, *D. arzelensis*, *D. bockhi*, *D. bulogensis*, *D. laluensis*, *D. udvariensis*, *Myophoria* cf. *goldfussi*, *M.* cf. *inaequicostata*, *Posidonia ovalis*, *Pleuronectites* cf. *difformis* and *Pteria* sp. found in the Baifeng Formation suggest that it is of Middle Triassic age [13]. The sediments commonly obtain 2000 m in thickness and represent terrigenous material (flysch) deposited upon Upper Devonian through Lower Triassic carbonates [13]. Although hiatuses are common throughout the Late Devonian through Middle Triassic sequence, significant angular unconformities are absent, indicating that the sediments were deposited in a broad ocean basin that was not significantly disturbed by tectonic activity during that time. The influx of terrigenous material in the Middle Triassic, as represented by the Baifeng Formation, suggests retreat of the sea and shrinkage of the ocean basin.

The Middle Triassic strata in Guangxi are mildly to intensely folded. The absence of Late Triassic through earliest Cenozoic sediments in the area makes it difficult to precisely determine the age of the folding. Jurassic through Cretaceous felsic intrusives emplaced during the Yanshanian orogeny are found in the area and could be responsible. Cenozoic deformation in the region resulting from the India–Asia collision has also been identified [14]. Fig. 3b is a simplified geologic map of the Tianyang area that includes the Yufeng section of western Guangxi. Because the strike of the Triassic folds parallels both the trend of the Youjiang fault and the strike of the Eocene sediments found in the fault zone, a similar age of deformation for all the sediments in the area is likely (i.e., Cenozoic). The folding in the Jinji area is more complex on a much smaller scale (Fig. 3c). Attitudes in the north of the section parallel the trend of the Youjiang fault while those in the south strike north.

We collected our samples using a portable drill, oriented them with a magnetic compass, and took sun compass bearings to correct for any local declination anomaly. We thermally demagnetized the samples in an oven built at the University of California at Santa Cruz that is housed in a shielded room. Calibration experiments show that the oven chamber is accurate to within $\pm 8°$ C of the set temperature; the ambient field in the cooling region is less than 10 nT. We measured the samples on a three-axis 2G superconducting magnetometer and analyzed the results using principal component analysis [15], the McFadden and McElhinny [16] great circle technique, and Fisher [17] statistics.

3. Paleomagnetic results

3.1. The Xikou Formation, central Fujian Province

Of the 4 localities sampled from the Xikou Formation, two (L152 and L153) contain samples

Fig. 4. Paleomagnetic data from the Xikou Formation, Fujian Province. Thermal vector demagnetization diagrams of sample 1591A in in-situ (a) and tilt-corrected (a') coordinates are shown. (b) and (c) Thermal vector demagnetization diagrams of two samples cut from the same core. One was demagnetized thermally (b) and the other with an alternating field (c). Note that the second component is better resolved when applying thermal demagnetization. ● = Vertical component; + = horizontal component. Temperature steps are in degrees Celsius; applied field steps are in milliTeslas; NRM = natural remanent magnetization. (d) and (e) Equal-area lower hemisphere projections in which 4 set points and 14 remagnetization great circles are fit to the data; (d) is in tilt-corrected coordinates. ☆ and ★ = Mean directions.

that yield northwest and up directions (in-situ coordinates) after removing a low-temperature component similar to the expected present field direction (Fig. 4a). We refer to the northwest and up direction as component two (C2) and the direction similar to the expected present field as component one (C1) ($D = 0.7°$, $I = 43.9°$, $\alpha_{95} = 4.7°$, $N = 3$ localities). Of the 19 samples from

Table 1
Summary of the paleomagnetic data from this study

Study/locality	Comp.	S/D	Coord	n/N	gc/sp	N/R	D(°)	I(°)	α_{95}	k
Fujian Province										
Xikou Fm.										
Mean	C2		G	4/19s	0/4	4/0	305.7	-62.7	11.0	70.4
Set Points			S	4/19s	0/4	4/0	301.5	-29.3	7.4	156.3
Mean	C2		G	13/19s	0/13	12/1	326.1	-60.3	10.1	17.7
(L152+L153)			S	13/19s	0/13	12/1	305.4	-27.4	6.9	36.9
Mean	C2		G	18/19s	14/4	17/1	305.5	-56.9	5.6	41.4
(L152+L153)			S	18/19s	14/4	17/1	299.0	-23.6	4.6	61.9
Mean	C2		G	18/19s	5/13	17/1	317.2	-58.4	8.7	16.8
(L152+L153)			S	18/19s	5/13	17/1	302.7	-25.2	6.0	34.2
Guangxi Province										
Yufeng Section										
L118	C2	137/61	G	9/9s	7/2	4/5	66.4	-49.0	15.3	13.9
			S	9/9s	7/2	4/5	59.9	9.8	15.3	13.9
L303	C2	112/50	G	7/7s	4/3	7/0	65.6	-19.6	24.1	7.9
			S	7/7s	4/3	7/0	65.1	17.9	24.1	7.9
L304	C2	141/49	G	9/9s	6/3	9/0	49.4	-30.8	8.9	38.4
			S	9/9s	6/3	9/0	49.5	18.1	8.9	38.4
L306	C2	307/68	G	10/10s	8/2	9/1	63.9	63.5	6.1	72.8
			S	10/10s	8/2	9/1	48.5	-1.9	6.1	72.8
L308	C2	311/58	G	6/6s	5/1	4/2	62.2	53.4	9.9	65.7
			S	6/6s	5/1	4/2	53.3	-2.6	9.9	65.7
L309	C2	326/68	G	7/7s	3/4	7/0	25.1	68.4	8.6	55.3
			S	7/7s	3/4	7/0	45.2	3.2	8.6	55.3
Mean	C2		G	15/64s	0/15	14/1	59.1	25.1	29.7	2.6
Set Points			S	15/64s	0/15	14/1	55.9	8.3	8.3	13.3
Mean	C2		G	6/8L			57.7	15.2	55.7	2.4
			S	6/8L			53.5	7.5	9.9	46.4
Guangxi Province										
Jinji Section										
Mean	C2		G	14/59s	3/11	14/0	51.3	71.3	4.0	100.2
			S	14/59s	3/11	14/0	71.7	3.0	3.6	122.3
Yufeng C2										
(6 localities)			G	7L			57.2	25.7	49.4	2.4
+ Jinji C2			S	7L			56.1	6.9	9.8	39.2

Comp. = Component; S/D = average strike and dip of locality (note that strikes and dips may vary within each locality; the dip direction is the strike plus 90°); *Coord.* = coordinate (G = in-situ (geographic) coordinates; S = tilt-corrected (stratigraphic) coordinates); n/N = number of samples used to calculate mean direction/number of samples demagnetized (s = samples; L = localities); gc/sp = number of great circles (gc) or set points (sp) used to define the mean direction; N/R = number of normal (N) and reversed (R) samples; D = declination; I = inclination; α_{95} = 95% confidence radius; k = best estimate of the precision parameter. The polarities for the Xikou Formation are assigned assuming the magnetization was acquired in the southern hemisphere.

localities L152 and L153, 4 decay linearly to the origin. In the latter samples, C1 is removed by 465–515° C, after which C2 persists until the magnetization is completely unblocked by about 580° C (Figs. 4a and a'). Alternating field (a.f.) demagnetization is also effective on these samples but the components are not isolated as well as when using thermal demagnetization (Figs. 4b and c). These demagnetization characteristics suggest that magnetite is the main carrier of the remanence in these samples.

In other samples from localities L152 and L153, C2 is clearly present but does not decay linearly to the origin. Three methods that use solely best-fit line segments [15] or combinations of best-fit line segments and great circles were used to analyze them; all of these methods yielded similar results (Table 1). The first method isolates C2 in 13 of the 19 samples by forcing and not forcing best-fit line segments to the origin depending on whether C2 decays univectorially to the origin. For this method, the best estimate of the precision parameter (k) increases 2.1 times, from 17.7 in in-situ to 36.9 in tilt-corrected coordinates, an increase in clustering that is significant at greater than 95% confidence [18]. In the second method, we fit great circles to the samples that do not decay linearly to the origin and use the ones that do decay linearly ($n = 4$) as set points [16] (Table 1 and Fig. 4d). In the third method, we again used the remagnetization great circle technique but used the line data defined by the first method ($n = 13$) as set points and combined them with the remainder of the samples ($n = 5$) that were fit with great circles (Table 1). Both techniques successfully isolate C2 in 18 out of the 19 samples from the two localities and demonstrate a passing of the fold test at greater than 95% confidence [19]. Note that this study is performed on the sample level, which precludes the application a fold test based on the degree to which the best estimate of the precision parameter (k) is maximized [20]. Because (1) the great circle method incorporates more samples than the principal component method, (2) set points are used to help define the mean directions and (3) the analytical methods yield similar directions, we prefer the results obtained by the great circle method. Of the two results obtained by the great circle method, the average number of demagnetization steps used to define a great circle is greater than that used to define a best-fit line. Thus, the mean direction calculated from 14 great circles and 4 set points is preferred. The dispersion of the sample directions before and after unfolding is shown in Fig. 4e.

The other two localities (L154 and L158) do not contain C2 as a part of their unblocking spectra. After the second demagnetization step (200° C), most samples from L158 behave chaotically, never defining a stable component. Conversely, the samples from L154 behave relatively stably but contain only C1 directions.

We note that our tilt-corrected direction isolated in this study is significantly different from that reported by Zhai et al. [21,22] ($D = 335.5°$, $I = 4.9°$, $\alpha_{95} = 7.9°$). Although the details of their study are unknown, a possible explanation for the discrepancy lies in Figs. 4b and c. The samples in the figures (1521A and B) come from the same core. These were analyzed using thermal and alternating field demagnetization techniques respectively. It appears that a greater percentage of the C2 component is unblocked during thermal demagnetization than it is by a.f. The component defined in the higher coercivity spectra from a.f. demagnetization yields a direction similar to that reported by Zhai et al. [21,22], which is less westerly and more equatorial than the direction isolated by thermal demagnetization as reported here. Thus, an incomplete separation of components may result in more northerly and shallower directions for the C2 component.

In sum, only samples collected from 2 of 4 localities possess a northwest and up direction, and of these samples only one is antipodal from the rest. However, these samples pass a fold test and some exhibit stable demagnetization paths that decay univectorially to the origin after removing an overprint. The successful fold test implies that the magnetization is pre-Late Cretaceous in age. Until Jurassic or Early Cretaceous paleomagnetic data become available to help constrain this result, we tentatively conclude that the C2 tilt-corrected direction (Table 1) is a primary remanence.

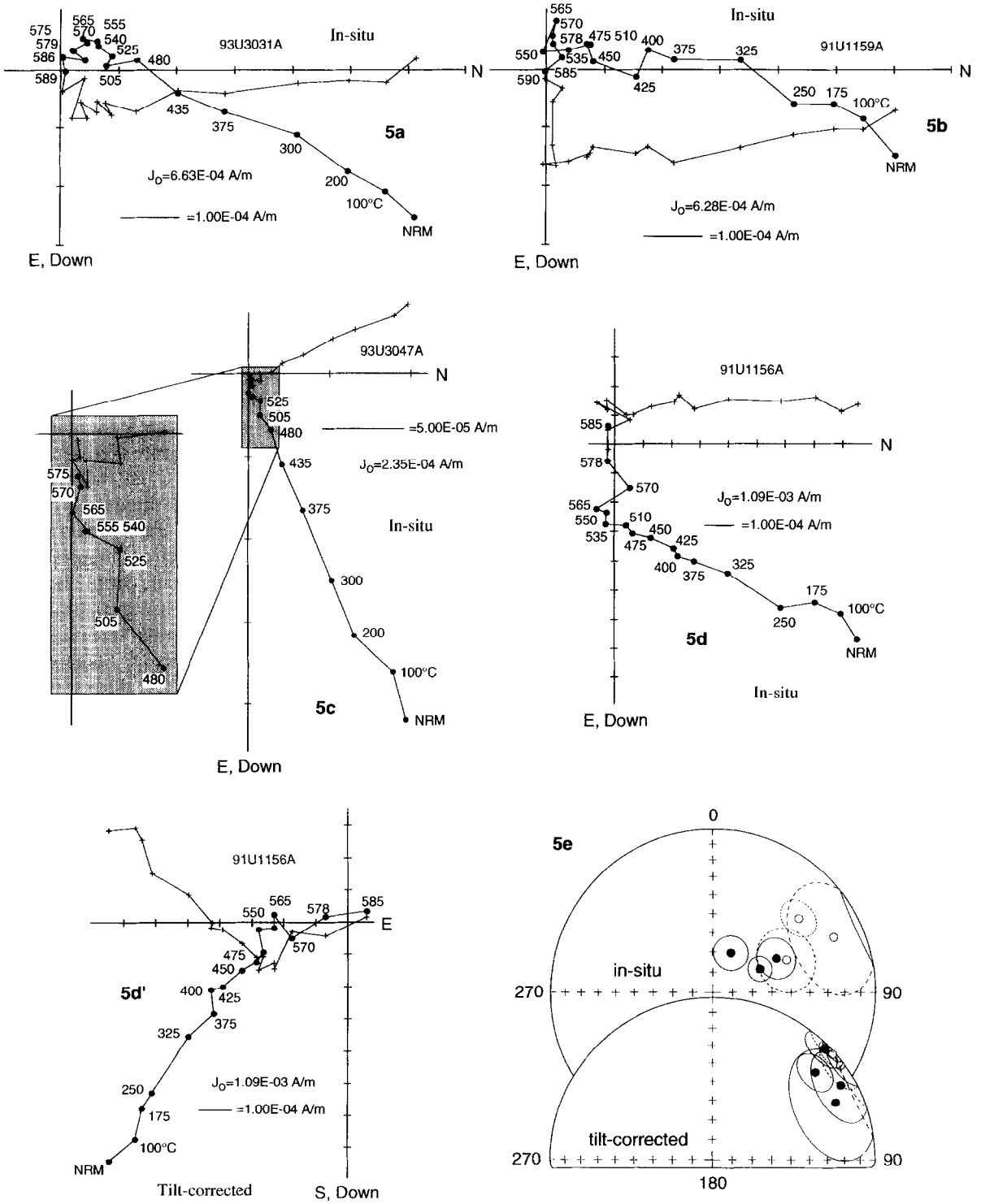


Fig. 5. Paleomagnetic data from the Baifeng Formation (Yufeng section), Guangxi Province. (a)–(d) Thermal vector demagnetization diagrams for samples from localities L304, L118, L306 and L118 respectively. Conventions are the same as those in Fig. 4. (e) Equal-area plot showing the mean directions with their 95% circles of confidence; dots (•) are on the upper hemisphere.

3.2. The Baifeng Formation (Yufeng Section), western Guangxi Province

The Yufeng section consists of a series of anticlines and synclines (Fig. 3b), from which we collected 64 samples at 8 localities. Most of the samples have weak (less than 10^{-3} A/m) intensities of natural remanent magnetization (NRM). They have unblocking temperatures ranging from 550 to 586° C and are readily demagnetized by a.f., both characteristics suggesting that the remanence is carried by magnetite. The NRM directions are quite variable between localities but the mean in-situ direction ($D = 358.8°$, $I = 38.7°$, $\alpha_{95} = 3.6°$, $N = 64$ samples) is similar to the expected present-day field direction and is probably a recent overprint.

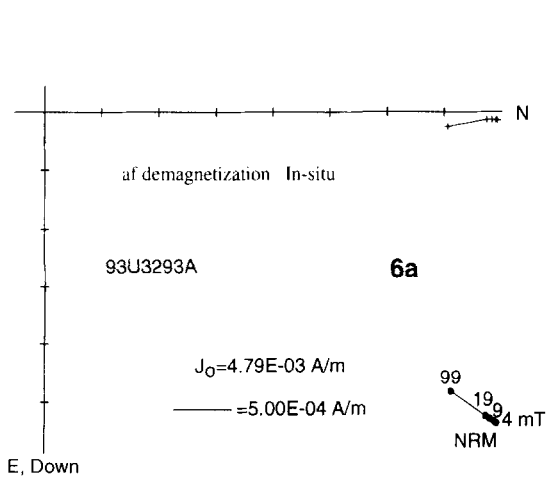
Although the magnetization is weak and generally unstable, a pattern emerges in the demagnetization paths after removal of the overprint. In in-situ coordinates, the progressive demagnetization directions of samples from localities with south-facing dips trend toward northeast and up (Figs. 5a and b), while those from localities with north-facing dips trend toward northeast and steeply down (Fig. 5c and Table 1). Samples with reversed directions (Figs. 5d and d′) are also identified but are few. By analyzing the data from each locality using a combination of remagnetization great circles with set points [16], we were able to isolate the above-mentioned directions in all samples from 6 of the 8 localities (Table 1). We omitted the other two localities from further analysis because in both cases the present field overprint persists and no other component is isolated.

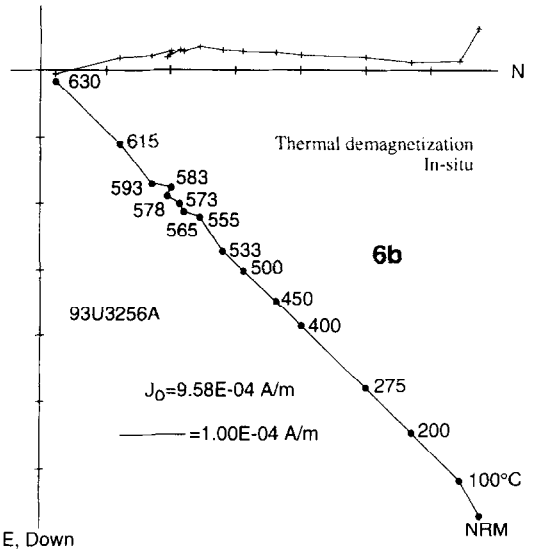
The deformed units in the Yufeng section provide a good opportunity to apply the fold test (Fig. 5e). For the 6 locality mean directions, the McFadden [19] fold test is positive at greater than 99% confidence at 83% unfolding (SCOS1 = 5.873 in-situ, 0.121 at 83% unfolding), and k reaches a maximum (93.7) at 86% unfolding. The tilt-corrected directions at 83% and 86% unfolding ($D = 53.3°$, $I = 8.8°$ and $D = 53.3°$, $I = 8.5°$ respectively) are nearly identical to that at 100% unfolding ($D = 53.5°$, $I = 7.5°$). The Watson and Enkin [20] fold test yields a similar conclusion by demonstrating that the optimum degree of untilting occurs at 87% unfolding. The discrepancy in the results is probably due to oversimplified bedding correction procedures (non-parallel fold axes (Fig. 3b), etc.) and not due to a synfolding magnetization, and we see no reason to prefer the other results over the one obtained at 100% unfolding.

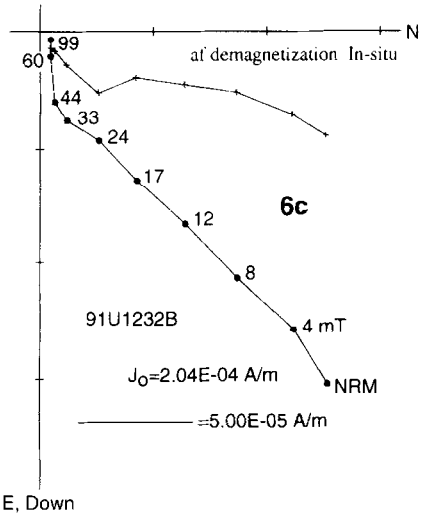
The positive fold test suggests the magnetization was acquired before folding, sometime in the Mesozoic or earliest Cenozoic. The mean tilt-corrected direction is distinct at more than 99% confidence from Early Jurassic and Early Cretaceous through Cenozoic directions from the area [14,23,24], suggesting that the magnetization is pre-Early Jurassic. Although few, the presence of samples with reversed polarity favors an interpretation of primary magnetization. Furthermore, the mean direction of these samples is antipodal to that of the samples with normal polarity within 95% confidence limits. We therefore regard the mean tilt-corrected direction to be a primary magnetization acquired in the Middle Triassic.

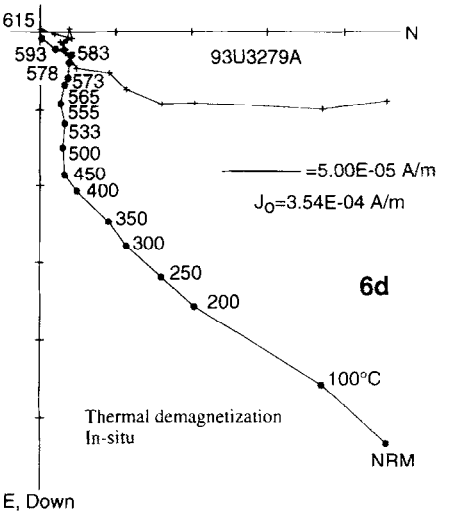
3.3. The Baifeng Formation (Jinji section), western Guangxi Province

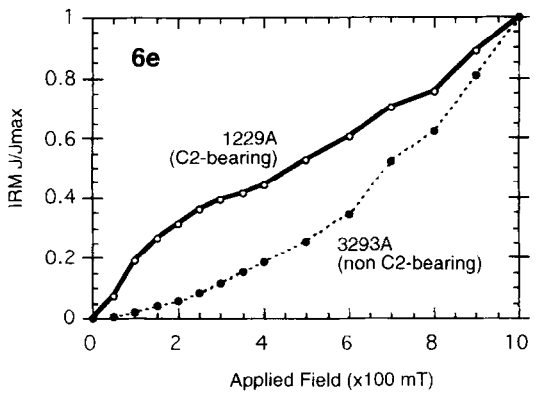
The Jinji section consists of very fine grained orange–red siltstone and mudstone. In the majority of the samples that we analyzed, alternating fields of 100 mT do not remove a significant percentage of the NRM, and thermal demagnetization unblocks most of the remanence between 600 and 640° C (Figs. 6a and b). The presence of high unblocking temperatures, the inability of an a.f. of 100 mT to remove a significant proportion of the magnetization, and the red color of the rocks all suggest that hematite is the main carrier of the remanence. These samples have very stable magnetizations that maintain north and down directions (in-situ coordinates) even at the higher unblocking temperatures (Fig. 6b). The north and down direction, referred to here as component one (C1), is isolated at all seven localities from the Jinji section, and k for the locality mean directions decreases drastically on passing from in-situ (413.8) to tilt-corrected coordinates (5.2); obviously, these samples fail a fold test. The similarity of the direction ($D = 1.9°$, $I = 36.8°$,


N
af demagnetization In-situ
93U3293A
6a
J_O=4.79E-03 A/m
=5.00E-04 A/m
99
19
9
4 mT
NRM
E, Down


N
630
615
Thermal demagnetization
In-situ
593
583
578
573
565
555
533
500
6b
450
93U3256A
400
J_O=9.58E-04 A/m
275
=1.00E-04 A/m
200
100°C
NRM
E, Down


N
99
60
af demagnetization In-situ
44
33
24
17
6c
12
8
91U1232B
4 mT
J_O=2.04E-04 A/m
NRM
=5.00E-05 A/m
E, Down


615
N
593
583
93U3279A
578
573
565
555
533
500
450
=5.00E-05 A/m
400
J_O=3.54E-04 A/m
350
300
250
6d
200
100°C
Thermal demagnetization
In-situ
NRM
E, Down


6e
1
0.8
0.6
0.4
0.2
0
IRM J/Jmax
1229A
(C2-bearing)
3293A
(non C2-bearing)
0
2
4
6
8
10
Applied Field (x100 mT)

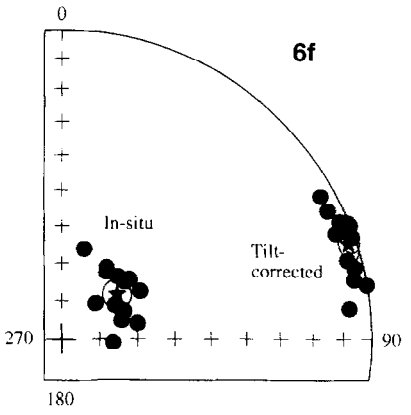

0
6f
In-situ
Tilt-
corrected
270
90
180

Table 2
Early and Middle Triassic paleomagnetic poles from south China

Province or Region	Location on Fig. 1	Lat.	Long.	Age	λ	ϕ	A_{95}	N	Reference
Hunan	2	29.5	110.0	T_{1-2}	37.1	197.8	-	2L	[28]
Sichuan	3	29.5	106.0	P_2 to T_2	40.7	215.0	6.8	5L	[27]
Guizhou	4	28.6	106.9	T_1	46.3	219.1	10.9	9L	[25]
Sichuan	5	32.0	105.5	P_2 to T_2	47.9	225.1	4.1	114s	[26]
Mean Yangtze	-	29.9	107.1	P_2 to T_2	43.5	213.6	11.4	4L	
Guangxi	1	23.6	107.2	T_2	32.2	208.7	7.9	7L	This study
Fujian	6	26.0	117.2	T_1	19.4	2.1	3.8	18s	This study
(Fujian adjusted	6	26.0	117.2	T_1	9.4	8.4	3.8	18s	This study)
North China	-	-	-	T_1	53.4	358.4	8.9	5L	[31]
North China	-	-	-	T_{2-3}	60.2	9.6	11.6	4L	[31]

Lat. = Latitude (° N) of locality; *Long.* = longitude (° E) of locality; *T* = Triassic; *P* = Permian; *1* = early; *2* = late; λ = latitude (° N) of pole; ϕ = longitude (° E) of pole; A_{95} = 95% confidence radius; *N* = number of localities (*L*) or samples (*s*) used to determine pole; *adjusted* = 12° of post-Cretaceous clockwise rotation removed. Note that the Hunan pole [28] is the average of the Daye and Jialingjiang results. The Fujian pole shown in Fig. 8b is the adjusted one.

$\alpha_{95} = 3.0°$, $N = 7$ localities) to that of the expected present field suggests that it is a recent overprint.

However, the samples from locality L127, two samples from locality L335 and two from locality L337 display demagnetization characteristics that differ much from those described above. In these samples, the overprint direction yields to a northeast and steeply down (in-situ coordinates) direction during a.f. and thermal demagnetization (Figs. 6c and d). A third component (C3) is then isolated after the 570–581° C temperature step using thermal demagnetization that is not unblocked by a.f. demagnetization. The C3 in-situ direction ($D = 2.6°$, $I = 39.7°$, $\alpha_{95} = 2.9°$, $N = 9$ samples, $k_g/k_s = 6.4$) is virtually identical to C1 and persists until complete unblocking occurs at 620–640° C.

The difference between the samples that have the second component (C2) from those that do not is that, for the former type, about 90% of the NRM is removed by 575° C and a.f. demagnetization is effective at removing a significant percentage of the original remanence. This suggests that C2-bearing samples possess magnetite as the dominant magnetic mineral. Isothermal remanent magnetization (IRM) experiments (Fig. 6e) also show differences in magnetic mineralogy between C2-bearing and non-C2-bearing samples, and do not preclude the presence of magnetite in the C2-bearing samples. The emergence of the third component after 575° C, the inability of a.f. demagnetization to resolve it, and the similarity in direction and uncommon unblocking temperature (620–640° C) compared to the overprinted samples described above, suggests that hematite is

Fig. 6. Paleomagnetic data from the Baifeng Formation (Yufeng section), Guangxi Province. (a)–(d) Vector demagnetization diagrams for samples from localities L337, L332, L127 and L127 respectively. Conventions are the same as those in Fig. 4. (e) Results from the isothermal remanent magnetization (IRM) experiments. (f) Part of an equal-area plot showing the sample directions before and after bedding correction. ★ = Mean directions.

also present in the C2-bearing samples. This is probably why C2 directions do not decay univectorially to the origin.

We analyzed the data from the Jinji section by identifying samples ($n = 11$) with well-defined linear C2 segments (e.g., Figs. 6c and d) and then applying principal component analysis [15]. We then used the line data as set points and combined them with great circles ($n = 3$) to define the mean in-situ and tilt-corrected directions [16]. Using this technique, C2 in each of the samples was isolated (Table 1), and the McFadden [19] fold test is positive at greater than 95% confidence, with definition two of the test statistic decreasing from 5.772 to 0.349 (Fig. 6f).

Like the result from the Yufeng section, the mean tilt-corrected direction from the Jinji section is distinct at greater than 99% confidence from all post-Triassic directions found in the region and is most likely representative of the Middle Triassic. After combining this result with the 6 locality mean directions from the Yufeng sec-

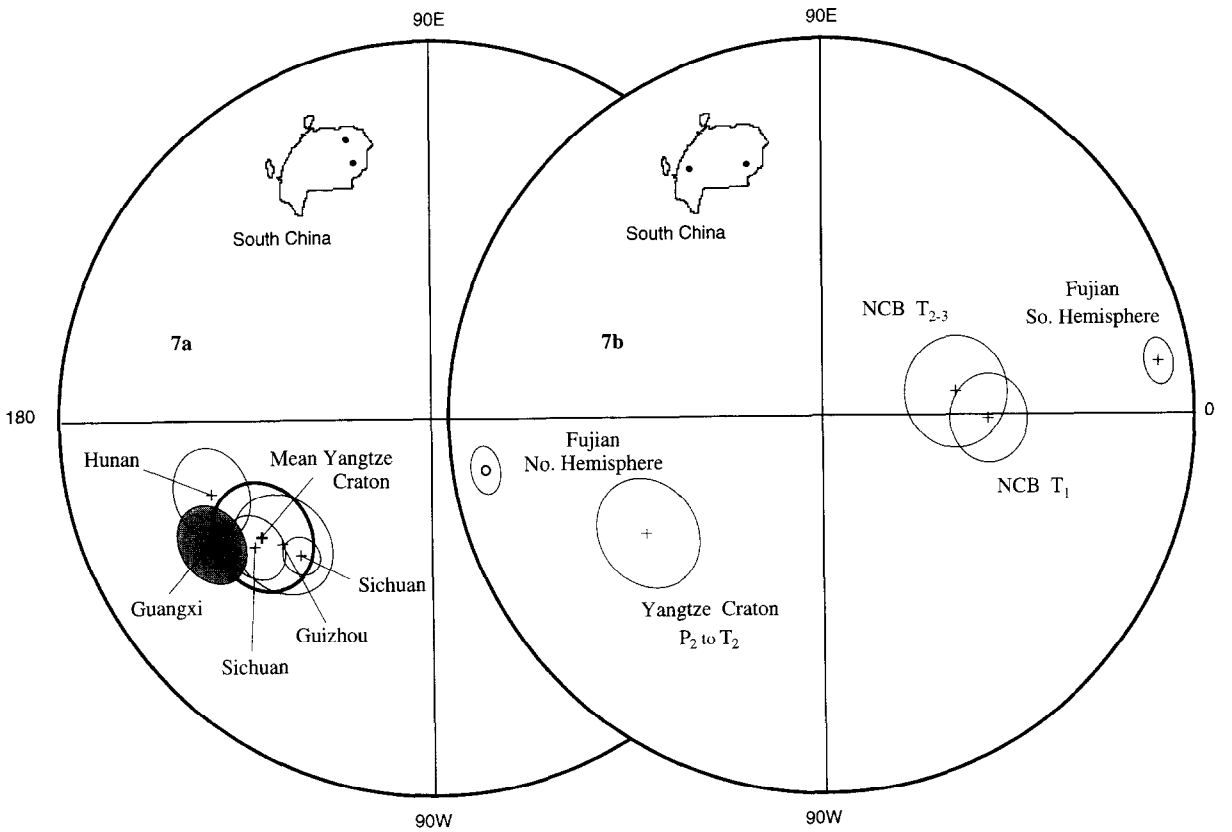


Fig. 7. Equal-area plots of (a) the Triassic paleomagnetic poles from the Yangtze craton and Guangxi Province and (b) the Middle through Early Triassic reference pole from the Yangtze craton, the Triassic reference poles from north China, and the pole from the Xikou Formation, Fujian Province (Table 2). For chronological abbreviatons, see Fig. 3. The shaded A_{95} circle in (a) is from the Middle Triassic Baifeng Formation from this study. The heavy A_{95} circle is the reference pole from the Yangtze craton and is an average of the four poles with the light A_{95} circles. Note that an A_{95} of 8.9° is arbitrarily assigned to the pole from Hunan province [28]. Both the pole from Guangxi and those from the Yangtze craton are well clustered, suggesting that localities 1 through 5 in Fig. 1 were part of the same continental landmass, at least by the Middle Triassic. In (b) the pole from the Xikou Formation (Fujian Province, 6 in Fig. 1) has been adjusted to remove a post-Cretaceous clockwise rotation (12°). With respect to the Yangtze craton, between the Early Triassic and the Late Cretaceous, Fujian may have been rotated 121 ± 9° counterclockwise and displaced 22 ± 9° north, or rotated 59 ± 9° clockwise and displaced 3 ± 9° south, depending on the hemisphere in which the magnetization was acquired; both possibilities are shown in the figure.

tion we find the mean direction for the Baifeng Formation to be $D = 56.1°$, $I = 6.9°$ ($\alpha_{95} = 9.8°$, $N = 7$) (Table 1).

4. Discussion

4.1. Reference Triassic paleomagnetic poles for the Yangtze craton

In order to apply these new results to tectonic questions, we first determined an Early through Middle Triassic reference pole for south China. The studies we used were based on the following criteria: (1) the magnetization of the rocks must be concluded to be primary and pass reasonable standards of reliability, (2) the rocks must come from the undisputed part of the Yangtze craton (e.g., Guizhou, northern Hunan and Sichuan provinces), and (3) the study must include rocks of Early or Middle Triassic age. Based in this, we averaged four poles [25] (Guizhou study only), [26–28] to define the reference pole (Table 2). Note that the results of Heller et al. [26] and Enkin et al. [27] supersede those of Jiang et al. [29] and Steiner et al. [30]. Also note that Table 2 uses an averaged pole from the Daye and Jialingjiang formations from Dobson and Heller [28]. Two of the poles [26,27] that we used to define our reference pole were derived from directions that span the Permo-Triassic boundary. The similarity of directions found throughout a Permo-Triassic section sampled by Steiner et al. [30] prompted those authors to conclude that little APW occurred from the Late Permian through the Middle Triassic in the Yangtze craton.

4.2. Tectonic quiescence in south China

The rocks studied here from the Baifeng Formation in Guangxi Province lie on what is referred to by Hsu et al. [4] as the Huanan Terrane (Fig. 1). According to these workers, the Baifeng sediments are a part of the Nanpanjiang accretionary wedge complex that was overthrust onto the Yangtze craton when the two blocks began colliding in the Triassic. The deformation found throughout the region is regarded by Hsu et al. as the product of the collision.

The paleomagnetic, stratigraphic and structural evidence provided here does not support this model. For example, the 4 poles used to define the reference pole, the mean of these poles, and the pole from the Baifeng Formation (Guangxi) are shown in Fig. 7a. It is apparent in this figure that the various poles are relatively well clustered. With respect to the reference pole, western Guangxi is insignificantly displaced $6 \pm 11°$ north and rotated $12 \pm 11°$ clockwise, suggesting that each of the study localities were a part of the same continent at least by the Middle Triassic. Given that (1) the latest Permian poles are indistinguishable from the Early through Middle Triassic poles, (2) the Triassic and Permian strata are correlated throughout western Guangxi, Guizhou, northern and western Hunan and Sichuan provinces (e.g., [32]) and (3) the Triassic units in these provinces do not exhibit angular unconformable relationships with underlying Permian strata [13,33–35], it is unlikely that suturing has occurred in this part of China since the Early Permian. Furthermore, since Early Cretaceous through Permian beds in Sichuan and Guizhou provinces are conformable [33,35], it appears that the region has experienced relative tectonic quiescence during that time. The deformation found in this area of south China thus seems best attributed to the India–Asia collision.

4.3. Tectonic terranes

Fig. 7b shows the mean Triassic poles for south China, the North China Block (NCB) [31] and Fujian (Table 2). Apparent in the figure is the great amount of relative counterclockwise rotation that the SCB must experience before docking with the NCB [2,36]. It is also apparent that the Fujian pole is significantly different from that of south China, revealing a scenario that differs much from the tectonic quiescence found farther west. Note that Fujian has rotated 12° clockwise with respect to south China since the Late Cretaceous [14], and in order to infer tectonic movement from the Triassic paleomagnetic results it is preferable to subtract this rotation

before comparing them. Table 2 therefore lists our observed and adjusted (with 12° of post-Cretaceous clockwise rotation removed) poles for the Xikou Formation. The pole shown in Fig. 7b is the adjusted one.

One problem in interpreting the discrepancy between the Fujian and south China poles is not knowing the hemisphere in which the magnetization from Fujian was acquired. If the hemisphere was the south one, the Xikou Formation has been rotated $121 \pm 9°$ counterclockwise and displaced $22 \pm 9°$ north with respect to south China. If it was north the Xikou has been rotated $59 \pm 9°$ clockwise and displaced insignificantly $3 \pm 9°$ south with respect to south China. Although the displacement is surprisingly large (2400 km), the counterclockwise sense of rotation and northward sense of displacement is consistent with a similarly rotated and displaced (although of lesser magnitude) Jurassic basin in southeastern Guangxi that was found to be rotated $20 \pm 6°$ counterclockwise and displaced $13 \pm 6°$ north with respect to the Yangtze craton [23]. Vertically dipping sediments lying in a fault zone that parallel this small basin on its northwest side were found to be overprinted. Because (1) the basin is rotated with respect to the Yangtze craton, (2) Jurassic sediments in the nearby basins are only gently folded and (3) the deformation appears to be localized in a narrow fault zone, post-Triassic tectonic models based on sinistral shear are preferred over those based on collision. Since the basin lies inboard of Fujian, it is likely that the tectonic regime affecting southeastern Guangxi also affected Fujian. Furthermore, counterclockwise rotation and northward displacement is consistent with structural kinematic indicators that support tectonic models based on sinistral shear and the lateral transport of terranes for this part of the circum-Pacific region (e.g., [12,37]). For these reasons we prefer the interpretation that the magnetization was acquired in the southern hemisphere. A more definitive interpretation should result when a more complete Mesozoic APWP is available for Fujian.

4.4. Where does Korea belong?

Similarities in rock types of coeval ages with seemingly parallel tectonic affinities have led some geologists to conclude that the Korean peninsula south of the Imjingang belt and west of the Honam fault is a part of the SCB (e.g., [38–41]) (Fig. 1). Using similar logic, these workers liken the geology outside this region to that of the Sino-Korean craton (NCB). To test this we compare the Mesozoic paleomagnetic data from the SCB with those from Korea (Table 3). The results included for comparison from Korea pass reliability tests (e.g., the fold test), with the exception of one which is the only Jurassic result available from Korea not lying in the rotated Ogcheon Zone. The magnetization of the rocks used in the Jurassic study is, however, assumed to be primary [43].

The Mesozoic poles are graphically portrayed in Fig. 8. The remarkable similarity of pole position through time is immediately apparent in the figure and suggests both places have behaved as a single tectonic block since the Triassic. This sur-

Table 3
Mesozoic paleomagnetic poles from the Yangtze craton and south Korea

Age	Yangtze Craton					Southern Korea				
	λ	ϕ	A_{95}	N	Reference	λ	ϕ	A_{95}	N	Reference
K_{1-2}	76.9	211.6	6.0	4L	[14]	68.0	205.0	6.0	3L	[42]
J_{1-2}	63.4	192.3	6.6	4L	[23]	58.9	203.4	14.3	7L	[43]
T_{1-2}	43.5	213.6	11.4	4L	This study	31.7	216.5	13.6	6L	[43,44]

K = Cretaceous; J = Jurassic; for other abbreviations, see Table 2. Note that the Cretaceous and Jurassic poles for the Yangtze craton represent averages of several studies like the one obtained for the Triassic used here. The Triassic pole from Korea is obtained by including the locality means given in Kim and Van der Voo [43] with the mean direction in Kim et al. [44]. Jurassic results from the rotated Ogcheon Block (Korea) are not included in this table.

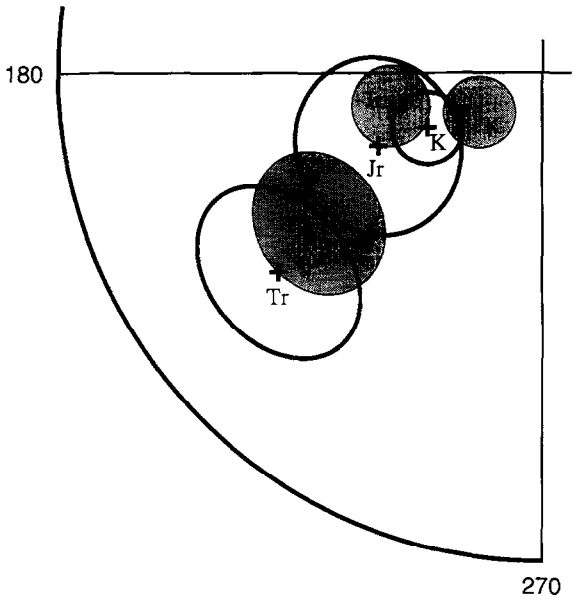


Fig. 8. Comparison of Mesozoic paleomagnetic poles from Korea (heavy circles) and the Yangtze craton (shaded circles) (data from Table 3). *K* = Cretaceous; *Jr* = Jurassic; *Tr* = Triassic. A remarkable coincidence of poles is observed, which, if true, implies that the South China Block (except for the displaced terranes in the coastal provinces) and Korea may have been part of the same continental landmass from the Triassic onwards.

prising coincidence, if true, has important implications for the tectonic development of this region of the circum-Pacific. Amongst other things it would suggest that transcurrent faults mapped in southeast China such as the Changle–Nan'ao fault are not related to the Tan-Lu system farther north (Fig. 1). If Korea was amalgamated to the SCB, these faults would lie on opposite sides of the large continental block (SCB + Korea), not dissecting it. This suggests that the Tan-Lu fault system would be related to the convergence and scissoring of the SCB (+Korea) with the NCB, and that motion along the system should terminate when the suturing was completed [45,46]. The abrupt termination of the Tan-Lu fault at the northern boundary of the SCB (e.g., [6]) supports this idea. Faults in southeast China would be influenced by proto-Pacific plate interaction. Temporal controls on these faults are unconstrained by this model but the conclusions based on the paleomagnetic data from Fujian are still valid. In closing it should be stressed that the Korean poles are only based on a few studies and more work is required to confirm the APWP for Korea presented here.

5. Conclusions

The magnetization in the Triassic rocks from central Fujian and western Guangxi Province is concluded to be of primary origin. However, limited numbers of samples in some cases and complex magnetic behavior in others should encourage further study. The paleomagnetic pole from western Guangxi is similar to other Triassic and Late Permian poles from field localities in Guizhou, Hunan and Sichuan provinces. This suggests that these areas were part of the same continent since the Triassic, and, in combination with geologic data, probably since the Late Permian. With respect to south China, Fujian could have been rotated $121 \pm 9°$ counterclockwise and displaced $22 \pm 9°$ northward, or rotated $59 \pm 9°$ clockwise and displaced $3 \pm 9°$ southward, depending on the hemisphere in which the magnetization was acquired. Based on consistency with other paleomagnetic results, and in accordance with the structural geologic data from the area, the former interpretation is preferred. Finally, the APWP from south Korea is compared to the south China APWP. A remarkable concordance of the paths suggests that these two areas have been part of the same tectonic unit since the Triassic. This observation, if true, implies that the major fault zones in north China (e.g., the Tan-Lu fault) are unrelated to major fault zones in southeast China (e.g., the Changle–Nan'ao fault), with the former probably associated with the suturing of north and south China and the latter influenced by proto-Pacific plate motion.

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